Queen bed frame

An Arts & Crafts queen bed frame in quartersawn white oak, made to wear a washable quilt over its headboard: square posts with pyramid caps, through-tenons pegged through the posts, wide flat slats to lean against, and a low foot rail with short capped legs.



Updated November 11, 2026. Online: https://github.com/sharpkathy123/Design-render/tree/main/deliverables/bed-frame

The design

64 7/8" wide over the tenons, 85 1/8" long, 60 1/8" to the post tops.

- **Headboard posts:** two 2 3/4" square posts with shallow pyramid caps. They stand 3 1/2" above the top rail, so they show above the folded quilt.
- **The headboard's rails** go right through the posts, with their tenons standing 3/8" proud on the outside faces. Each tenon is held by two square oak pegs through the posts' front faces. This through-tenon joinery is the most recognizable mark of Stickley's furniture.
- **Seven wide flat slats** sit flush with the rails' faces, so you lean against a flat panel.
- **The top rail** is at 56", 32" above the mattress top, so someone tall can sit up against it with pillows (raised from 49", 2026-11-10). It has well-rounded 3/8" edges.
- **Foot:** no tall posts. A straight foot rail keeps the mattress in place (an arched one showed the support under the bed, dropped 2026-11-11). It's through-tenoned and pegged into two short legs. The legs have the same 2 3/4" section as the head posts and stand 1" above the rail with the same pyramid cap, as period beds' low posts did. Their tops are still 6" below the mattress top (decided 2026-11-10: hiding them under the rail would be a modern platform-bed look).
- **Side rails:** on one bed bolt and a stub tenon at each end, the way period beds were. The foot's bolts have aged-brass covers, the small round caps on the foot legs (darkened 2026-11-11 so they don't stand out).
- **Support:** 2x4 cleats inside the side rails, thirteen poplar slats, and a 2x4 center rail on three legs stained dark (they can be seen under the foot rail from low down).
- **Finish:** fumed, a deep brown, as Stickley's workshops did it; or dyed to match (BUILD.md, step 15).



The same, from above the foot



With a quilt draped over the headboard



The bare frame



From the foot

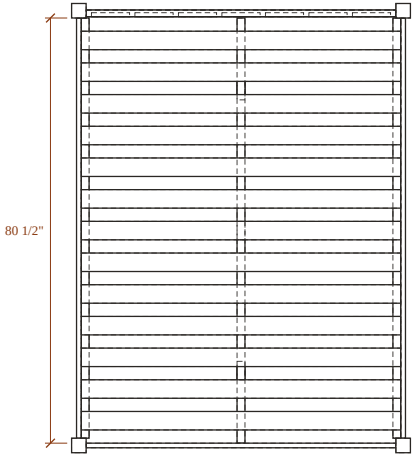


A head post's pyramid cap, and the top rail's through-tenon and pegs

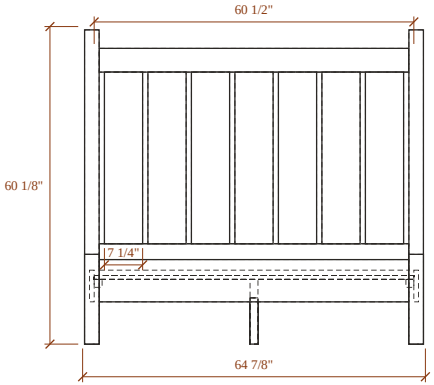


The foot's corner, with its pegged through-tenon and brass bolt cover

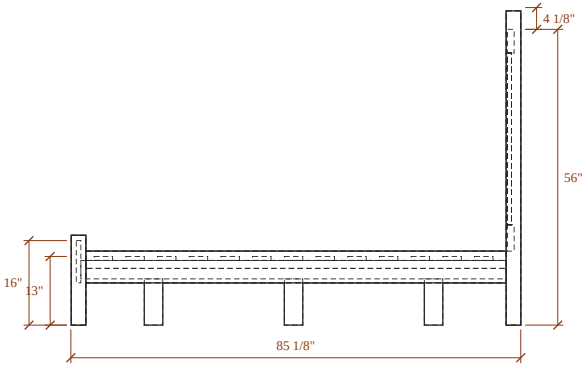
Overall drawing and cut list



TOP



FRONT



RIGHT SIDE

Queen bed frame

64 7/8" wide x 85 1/8" deep x 60 1/8" high, in inches

12"

CUT LIST

Qty	Part	Material	T x W x L
2	Head post	QS white oak	2 3/4" x 2 3/4" x 60 1/8" with its cap
1	Headboard bottom rail	QS white oak	1 1/4" x 5" x 64 7/8" with tenons
1	Headboard top rail	QS white oak	1 1/4" x 4 1/2" x 64 7/8" with tenons
7	Headboard slat	QS white oak	3/4" x 7 1/4" x 34" with tenons
2	Foot leg	QS white oak	2 3/4" x 2 3/4" x 17 5/8" with its cap
1	Foot rail	QS white oak	7/8" x 8" x 64 7/8" with tenons
2	Side rail	QS white oak	7/8" x 6" x 80 9/16" with stub tenons
2	Cleat	2x4 (SPF)	1 1/2" x 1 1/2" x 79 9/16"
13	Slat	poplar	3/4" x 3 1/2" x 60 1/2"
1	Center rail	2x4 (SPF)	1 1/2" x 3 1/2" x 80 1/2"
3	Center leg	2x4 (SPF), stained dark	1 1/2" x 3 1/2" x 8 3/4"

QS white oak: 31.5 board feet net (buy about 43 with waste)
2x4 (SPF): 5.4 board feet net (buy about 8 with waste)
poplar: 14.3 board feet net (buy about 20 with waste)
2x4 (SPF), stained dark: 1.0 board feet net (buy about 2 with waste)

Quartersawn white oak, fumed; poplar slats; 2x4 cleats and center rail. The headboard's rails and the foot rail are through-tenoned and pegged; side rails on bed bolts. Joinery: bed-frame-joinery.svg.
Mattress 60" x 80" x 11" on the slats (no box spring); its top 24" from the floor.

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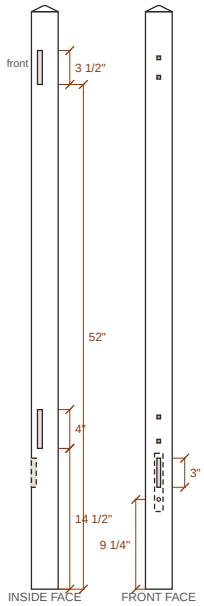
Joinery

Queen bed frame: joinery

Inches. Heights from the floor; post and leg positions from their front faces (toward the foot). Details drawn to different scales: work from the numbers. Overall plan and cut list: bed-frame-drawing.svg.

A. HEAD POST (2)

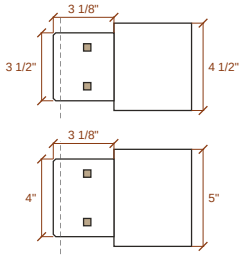
2 3/4" square, 60 1/8" long with its cap.
Left post shown; the right is its mirror.



Through-mortises 1/2" x 3 1/2" and 1/2" x 4", 5/8" from the front face, cut halfway from the inside and outside faces so neither blows out.
Pegs 3/8" square, 1 1/2" deep from the front face, centered across it, at 15 1/4", 17 3/4", 52 3/4", 54 3/4" up.
Stub mortise for the side rail: 3/8" x 3" x 9/16" deep, centered, 10 1/2" up.
Bed bolt: 3/8" hole through, front to back, 9 1/4" up, centered;
1" counterbore 1/2" deep on the back face for the bolt head.
Pyramid cap: 5/8" rise to a 1/4" flat; plane or sand it before glue-up.

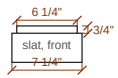
B. HEADBOARD RAILS: THROUGH-TENONS

1/2" thick tenons, centered on the 1 1/4" rail; shoulders 1/2" top and bottom.
The dashed line is the post's outside face: the tenon stands 3/8" proud, its end chamfered 1/8".



C. HEADBOARD SLATS (7) AND THEIR MORTISES

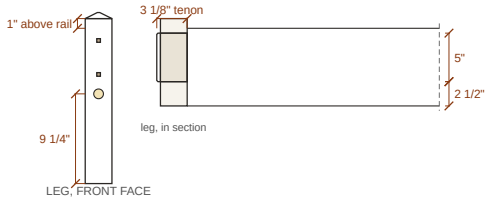
Slats 3/4" x 7 1/4" x 34" with tenons (32 1/2" between the rails).
Stub tenons 3/8" thick x 3/4" long, 1/2" shoulders each side; flush with the rails' front faces.



Mortises in both rails, 3/8" wide x 13/16" deep, 3/16" from the rails' front faces; measured from the rail's shoulder (the post's inside face):
slat 1: 1 1/2" to 7 3/4"
slat 2: 9 3/4" to 15 15/16"
slat 3: 17 15/16" to 24 3/16"
slat 4: 26 3/16" to 32 7/16"
slat 5: 34 7/16" to 40 11/16"
slat 6: 42 11/16" to 48 7/8"
slat 7: 50 7/8" to 57 1/8"

D. FOOT LEGS (2) AND FOOT RAIL

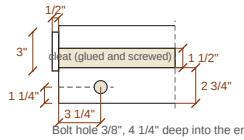
Legs 2 3/4" square x 17 5/8" with the cap; the rail 7/8" thick, 8" tall at its ends, straight.
Left leg and the rail's left end shown; the right is the mirror.



Rail 58 5/8" long between the legs, 64 7/8" overall with its tenons.
Through-tenons 3/8" thick, centered on the rail; mortises 1 3/16" from the legs' front faces.
The rail is straight; ease its edges 1/16".
Bed bolt from the front face, 9 1/4" up, 1" counterbore 1/2" deep, brass cover;
stub mortise for the side rail on the back face, as on the head posts.
Pegs 3/8" square through the front face, centered, at 11 1/4" and 14 3/4" up.

E. SIDE RAILS (2): ENDS

7/8" x 6", 79 9/16" between the posts and legs, 80 9/16" overall with the stub tenons.
The inside face shown (the end at a post: both ends the same).



Bolt hole 3/8", 4 1/4" deep into the end grain, 1 1/4" up, centered.
Nut hole 1" Forstner from the inside face, 3/4" deep, 3 1/4" from the end:
check it against the bolt before drilling.
Stub tenon 3/8" thick, 3" tall, 1/2" long, centered.
Cleats run the rail's full length, above the nut hole; each slat screwed to them with one #8 x 1 1/4" at each end, and to the center rail with one.

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How to build it

Everything to build it: what to buy and roughly what it costs, the tools, and the steps in order. The sizes are on two drawings:

- bed-frame-drawing.svg: the front, side and top views, the overall sizes and the cut list (finished sizes, tenons included)
- bed-frame-joinery.svg: every mortise, tenon, peg and bolt hole, dimensioned for marking out

Every number comes from sizes.py; change a size there and rerun the model and both drawings, never on the drawing alone.

What to buy, and roughly what it costs

Prices are estimates from November 2026. The lumber prices come from three suppliers' online lists (sources at the end); the hardware and finishing prices are typical retail and weren't checked against a store. Your local yard will differ, so take this list in and ask for a quote.

Lumber

Rough lumber, with about 20% extra for defects, planing and the odd bad board. Ask for **quartersawn** white oak (straight grain, ray fleck on the wide faces), FAS or Select grade.

Stock	For	Rough sizes to look for	Board feet	Price per bf	Cost
QS white oak 6/4	head posts and foot legs (each glued up from two boards), headboard rails	3 1/4" wide x 64" (4), 3 1/4" x 20" (4), 5" and 5 1/2" x 67"	about 21	$15 to $17 (estimated: 6/4 isn't on the lists checked)	$315 to $355
QS white oak 5/4	side rails, foot rail	6 1/2" x 82" (2), 8 1/2" x 67"	about 16	$14 to $16 (estimated, between 4/4 and 8/4)	$225 to $255
QS white oak 4/4	headboard slats	7 3/4" x 36" (7), or wider boards ripped	about 16	$13 to $14.50	$210 to $235
Poplar 4/4	13 mattress slats	4 1/4" x 62" (13)	about 26	about $5.35	about $140
2x4 x 8', kiln-dried, straight	center rail, its 3 legs, 2 cleats ripped from one		3 boards	$4 to $6 each	$12 to $18
Lumber					**about $900 to $1,000**

- **No jointer or planer?** Most yards will plane and straight-line rip for roughly $0.50 to $1 per board foot ($30 to $60 here). Give them the finished thicknesses: posts' boards 1 3/8", headboard rails 1 1/4", side and foot rails 7/8", slats 3/4".

- **Cheaper slats:** home-center 1x4 pine instead of poplar, about $8 per 8' board, one slat each (about $105). No one sees them.

Hardware

Item	Qty	Each	Cost
Bed bolts, 3/8" x 6", with square nut and washer	4	$4 to $6	$16 to $24
Brass bed-bolt covers, 1 5/8" (the two at the foot show)	2	about $6.40	about $13
Wood glue, Titebond III, 16 oz	1		$10 to $12
#8 x 1 1/4" wood screws (slats), box of 100	1		$8 to $10
#10 x 2 1/2" wood screws (cleats, center legs), about 40	1 box		$8 to $10
Felt or nylon glides for 7 feet	1 pack		$8 to $12
Sandpaper, 120, 150, 180			about $20
Hardware			**about $85 to $100**

The 3/8" square pegs are made from oak offcuts (step 9).

Finish: choose one

Fumed (the Stickley workshops' way: ammonia fumes darken the oak's own tannins, so the color is in the wood, not on it):

Item	Cost
Strong ammonia, 26% to 29%, 1 gallon (blueprint or chemical suppliers, not household ammonia)	$40 to $60
Plastic sheeting and tape for the tent	about $15
Respirator with ammonia cartridges, chemical goggles, gloves	$40 to $60

Or dyed (easier and safer, nearly the same look): a medium-brown water- or alcohol-based wood dye, about $20 to $25.

Then, either way: boiled linseed oil or an oil-varnish blend ($15 to $20), amber or garnet shellac ($20 to $30), dark paste wax (about $15).

Finish	Cost
Fumed, with the topcoats	about $150 to $200
Dyed, with the topcoats	about $75 to $90

Total

	Low	High
Lumber	$900	$1,000
Hardware	$85	$100
Finish (dyed / fumed)	$75	$200
Total	**about $1,050**	**about $1,300**

Add $30 to $60 if the yard planes the lumber.

Tools

- **Must have:** table saw (or a circular saw with a straightedge), drill and a 3/8" brad-point bit at least 8" long, 1" Forstner bit, chisels (3/8", 1/2", 1"), a mallet, a combination square and marking gauge, a jigsaw, a router with a 3/8" roundover bit, a random-orbit sander.
- **Clamps:** four that open to 72" (pipe clamps are cheapest) for the headboard, and six of 24" or more.
- **Makes it much easier:** a jointer and planer, a drill press or a doweling guide to bore the bolt holes straight, a hollow-chisel mortiser (or rent time at a makerspace).

The steps

1. **Let the wood settle.** Stack it with spacers in the room where you'll work for a week or two before milling.
2. **Mill.** Flatten and plane every board to its thickness and rip it a little wide; cut everything 1" long. Pick the best quartersawn faces for the posts' fronts and outsides, the slats' fronts and the foot rail's front.
3. **Glue up the posts and legs.** Two 1 3/8" boards for each, quartersawn faces out. When the glue is dry, plane to 2 3/4" square. Each post then has two quartersawn faces and two plainer ones: put the plain ones at the back and on the inside. (Stickley glued thin quartersawn faces onto the plain sides, so all four faces showed the figure. You can do that with 1/8" strips if you want it.)
4. **Cut to length and shape the caps.** Head posts 60 1/8", foot legs 17 5/8". Plane or sand the pyramid caps on all four, head posts and foot legs alike: 5/8" rise to a 1/4" flat, all four sides the same. Ease all the long edges 1/16".
5. **Lay out every mortise** from the joinery sheet, all at once, from the floor end and the front face (sheet A for the posts, D for the legs, C for the slat mortises in the rails).
    - Cut the **through-mortises** halfway from each side so neither face splinters.
    - Cut the **stub mortises** for the side rails.
    - Cut the **slat mortises** in both headboard rails.
6. **Cut the tenons** (sheets B, C, D, E).
    - Leave the through-tenons 3/8" proud, chamfer their ends 1/8", and sand them smooth: they're on show.
    - Fit each tenon to its own mortise and mark the pair.
7. **The foot rail** stays straight (an arch was tried and dropped: it showed the support under the bed). Ease its edges 1/16" like the rest.
8. **Round the headboard's top rail.** Use a 3/8" roundover on all four long edges, which is soft to lean on and easy on the quilt. Dry-fit the whole headboard and the foot. Check that the headboard is square and flat, and that the slats sit flush with the rails' faces.
9. **Pegs.**
    - Rip 3/8" square sticks from straight-grained oak offcuts and taper the lead end slightly.

    - Bore the peg holes through the posts' and legs' front faces with the joints dry-fitted.
    - Optional, for a joint that pulls itself tight: take the tenons out, and bore their holes 1/32" closer to the shoulder ("drawboring").
10. **Bed-bolt holes.**
    - Bore the 1" counterbore, 1/2" deep, then the 3/8" hole straight through. Bore from the back face on the head posts and from the front face on the foot legs.
    - Clamp each side rail in place, its stub tenon in its mortise. Run the long bit back through the post's hole into the rail's end, to 4 1/4" deep.
    - Drill the 1" nut hole from the rail's inside face, 3/4" deep. Test the bolt before moving on.
11. **Sand** to 150, then 180. Raise the grain with a damp cloth and sand again lightly if you'll use a water-based dye.
12. **Glue up the headboard.**
    - Glue only the middle of each slat's tenons, so the slats can move a little. Glue the rails' tenons into the posts.
    - Drive the pegs with a little glue, and trim them 1/16" proud with chamfered edges.
    - Clamp, check the diagonals for square, and clean off every bit of squeeze-out: dried glue won't take a fumed or dyed color.
13. **Glue up the foot.** The foot rail goes into the two legs the same way, with pegs.
14. **Cleats.** Glue and screw a 1 1/2" square cleat (ripped from a 2x4) along each side rail's inside face, its top 4 1/4" above the rail's bottom edge. Use #10 x 2 1/2" screws every 8".
15. **Finish.**
    - **Fumed:** fume outdoors or in a garage, never indoors. Set the parts in a plastic tent with a shallow dish of the ammonia and wear the respirator, goggles and gloves. Fume test offcuts first, checking every few hours. Most quartersawn oak goes a medium brown in 12 to 24 hours, and pieces from different boards darken differently.
    - **Dyed:** follow the dye's directions and match it on offcuts first.
    - **Then:** a coat of oil, two or three of shellac, and dark paste wax buffed out.
    - Stain the three center-rail legs dark: from low down they can be seen under the foot rail.
16. **Set it up in the bedroom.**
    - Bolt the side rails to the headboard and the foot, with brass covers on the foot bolts.
    - Set the center rail on its three legs down the middle and adjust its glides so it touches the slats' undersides.
    - Lay the 13 slats about 2 1/2" apart. Screw each one to the cleats and the center rail.
    - Put glides on all the feet.

Sources for the prices

- The Wood Yard, current price list (quartersawn white oak 4/4, poplar)
- Woodshed Lumber, white oak (quartersawn 4/4 and 8/4)
- Advantage Lumber, quartersawn white oak
- Lee Valley, brass bed-bolt cover
- Rockler, 3/8" bed bolt (price not shown on the page checked)